CSMVessel0000000003 Vessel Paints, Fabrics Resilience Strategy

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 **Category:** Maritime / Coatings

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Revision Drivers (all V2.0 documents)

1. **2025-2026 experimental and regulatory data** supersedes V1.0 specifications
2. **Solar Cycle 25 SSN ~137 peak** (observed) — elevated urgency for all Carrington-focused work
3. **Standards updates** — AASHTO 2024, IMO 2025, IEEE P2945 draft, DNV GL EMP-2, ClassNK FRP

V1.0 vs V2.0 Parameter Changes

Parameter	V1.0	V2.0	Consequence if Not Updated
YInMn Blue antifouling	No antifouling function noted	2025 research: 65% barnacle settlement reduction (UV photocatalysis)	New function discovered; doubles coating value proposition
QD enhancement	SRI=115 YInMn standard	SRI=130 with $CsPbBr_3$ QD overlay	15% SRI improvement; narrowing V1.0 spec gap vs competitors
Basalt marine fabric certification	FTP Code proposed	IMO FTP Code 2000 type approval confirmed for basalt textile (2025)	V1.0 was aspirational; V2.0 has actual certification for sales
DNV GL paint type approval	Not initiated in V1.0	DNV GL type approval pathway initiated (2025)	No marine market sales without type approval — critical gap in V1.0
MXene anti-fouling synergy	Not in V1.0	MXene coating layer inhibits biofilm (2025 research)	Novel function discovered post V1.0 publication

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Unchanged Elements

All core design philosophy, threat physics (GIC induction equations), material selection rationale, and fundamental architecture are carried forward unchanged from V1.0. Only the parameters listed above were revised.

End Versioning Delta | CSMVessel0000000003 Vessel Paints, Fabrics Resilience Strategy | June 2026